Introduction

We have studied the elementary probability in the previous chapter. As an application of probability, there are two more concepts namely random variables and probability distributions. Before seeing the definition of probability distribution, random variable needs to be explained. It has been a general notion that if an experiment is repeated under identical conditions, values of the variable so obtained would be similar. However, there are situations where these observations vary even though the experiment is repeated under identical conditions. As the result, the outcomes of the variable are unpredictable and the experiments become random.

We have already learnt about random experiments and formation of sample spaces. In a random experiment, we are more interested in, x number associated with the outcomes in the sample space rather than the individual outcomes. These numbers vary with different outcomes of the experiment. Hence it is a variable. That is, this value is associated with the outcome of the random experiment. To deal with such situation we need a special type of variable called random variable.

9.1 Definition of random variable

Definition

Let S be the sample space of a random experiment. A rule that assigns a single real number to each outcome (sample point) of the random experiment is called random variable.

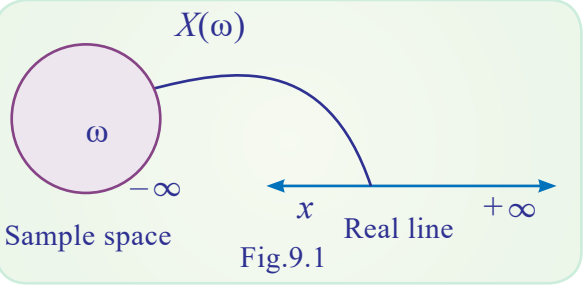


Fig.9.1

In other words, a random variable is a real valued function defined on a sample space S that is with each outcome ω of a random experiment there corresponds a unique real value x known as a value of the random variable X. That is $X(\omega) = x$.

Generally random variables are denoted by upper case alphabets like X, Y, Z … and their values or realizations are denoted by the corresponding lower case letters. For example, if X is a random variable, the realizations are $x_1, x_2 \ldots$

For example,

(i) Consider the random experiment of rolling a die.

The sample space of the experiment is $S=\{1, 2, 3, 4, 5, 6\}$

Let X denotes the face of the die appears on top.

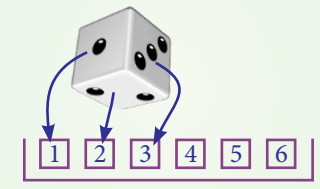


Fig.9.2